A Generalization Strategy for Speech Quality Prediction: From Domain-Specific to Unified Datasets

Imran E Kibria, Ada Lamba, Donald S. Williamson

kibria.5@osu.edu, lambda.39@osu.edu, williamson.413@osu.edu

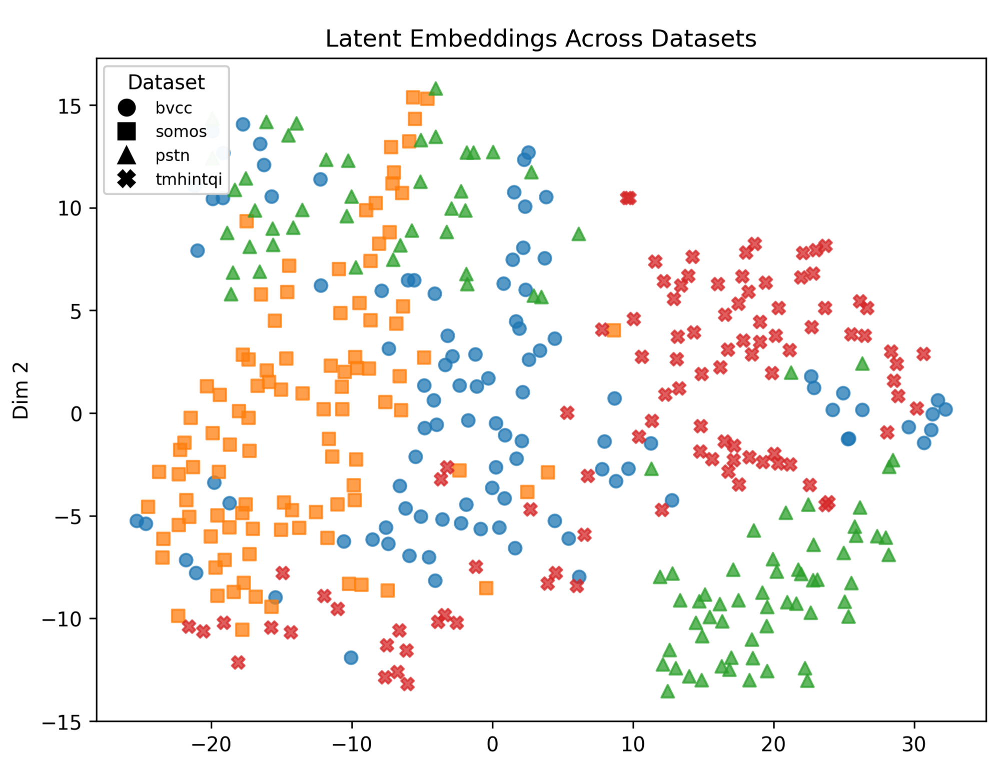
1. MOTIVATION

Objective

Predict human-perceived speech quality (MOS).

Challenges

- MOS prediction is dataset dependent.
- Models exhibit poor domain generalization.
- Domain diversity arises from differences in:
  - Language
  - Sampling rate
  - Speech type (natural, synthetic)



Research Problem

How can we train models that generalize across domains/datasets?

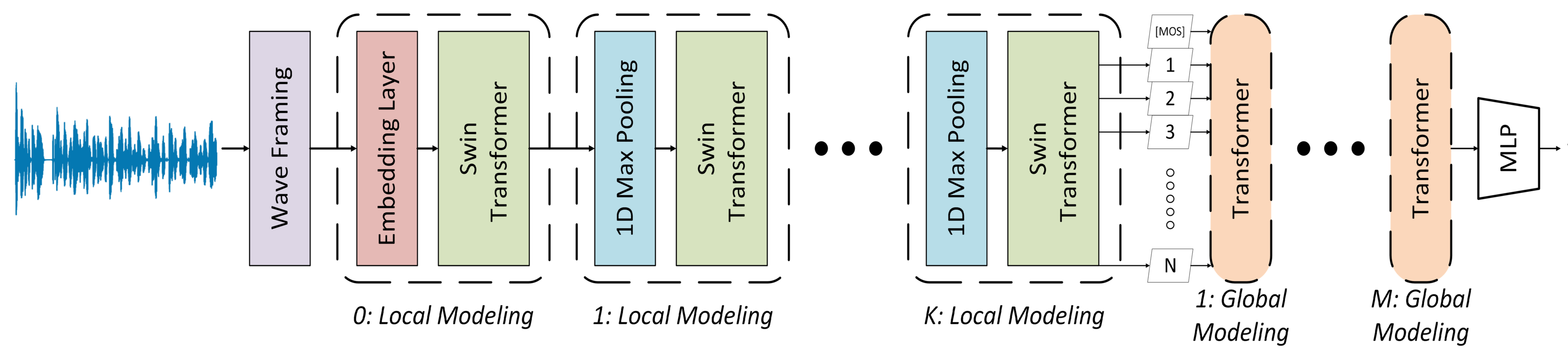
Proposed Solution

SAM enables better domain generalization when training on unified datasets.

4. LIMITATIONS

1. Challenging performance on extreme out-of-domain datasets.
2. Limited robustness to unseen languages (e.g., German, French).
3. Certain datasets need further analysis (NISQA FOR)

2. METHOD



Model: AttentiveMOS

[Kibria & Williamson, Interspeech 2025]

- Lightweight (86K parameters)
- Attention-only architecture
- Input: waveform → frame embeddings → transformer → linear head → MOS

Training Setup

- 7 training sets (individual + unified)
- 12 evaluation sets (seen + unseen)
- Compare: Adam vs. SAM + Adam

Sharpness Aware Minimization (SAM)

[Foret et al., ICLR 2021]

Two-step update:

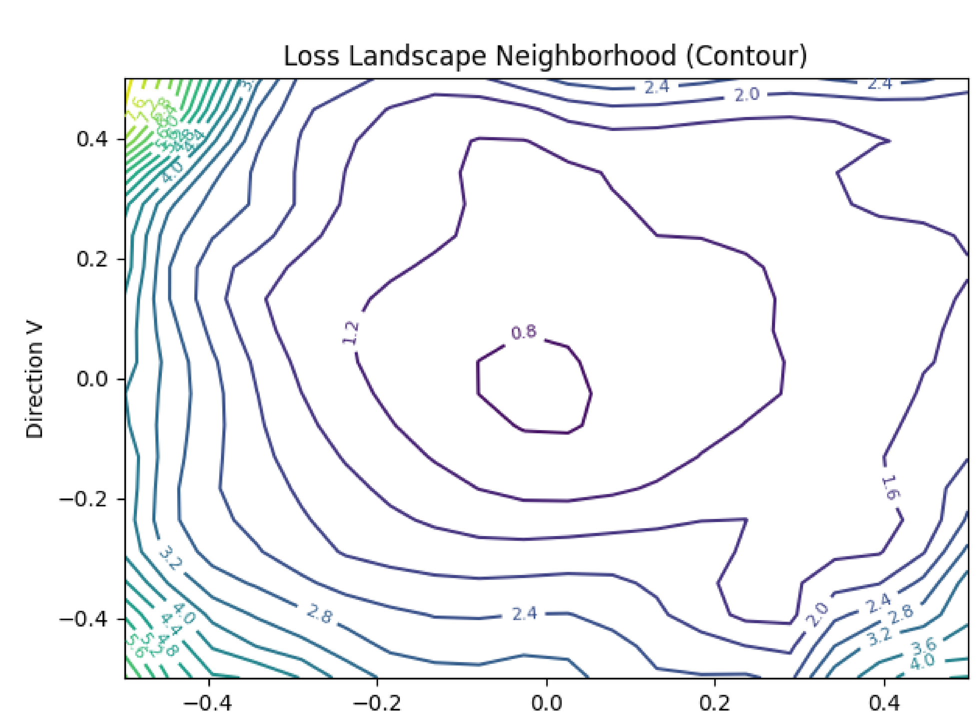
Step 1: maximize neighborhood loss

$$\widetilde{w} \stackrel{\text{def}}{=} w + \rho \frac{\nabla_w L(\boldsymbol{w})}{\| \nabla_w L(\boldsymbol{w}) \|_2}$$

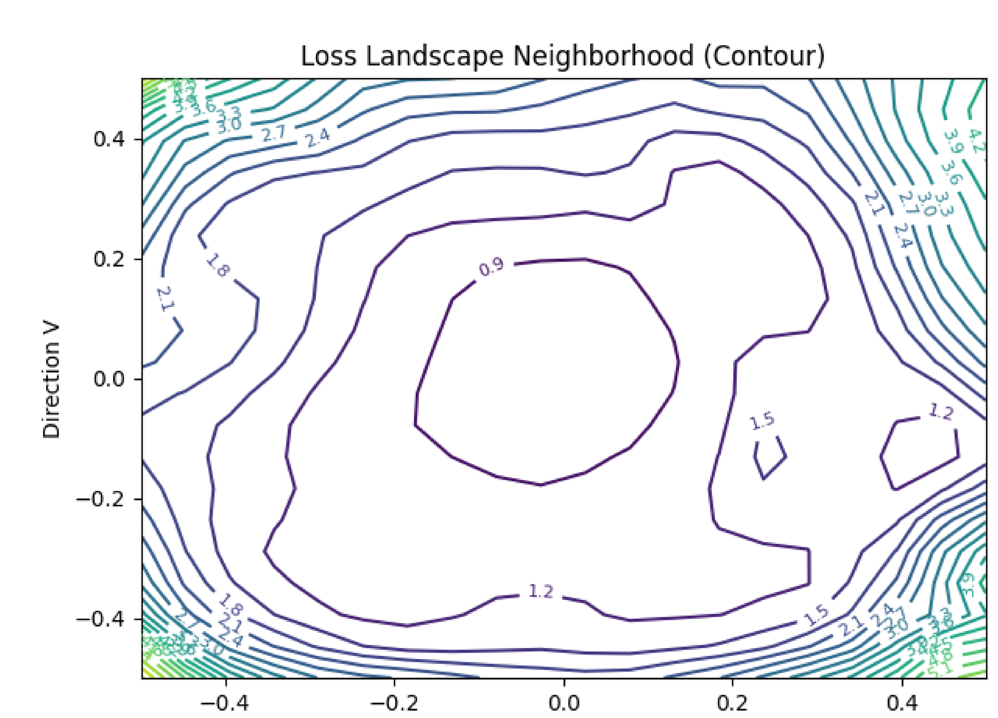
Step 2: minimize worst-case loss

$$w \leftarrow w - \eta \frac{1}{B} \sum_{i=1}^{B} \nabla_w L(\widetilde{w})$$

Model Flatness with AdamW



Model Flatness with SAM + AdamW



3. RESULTS

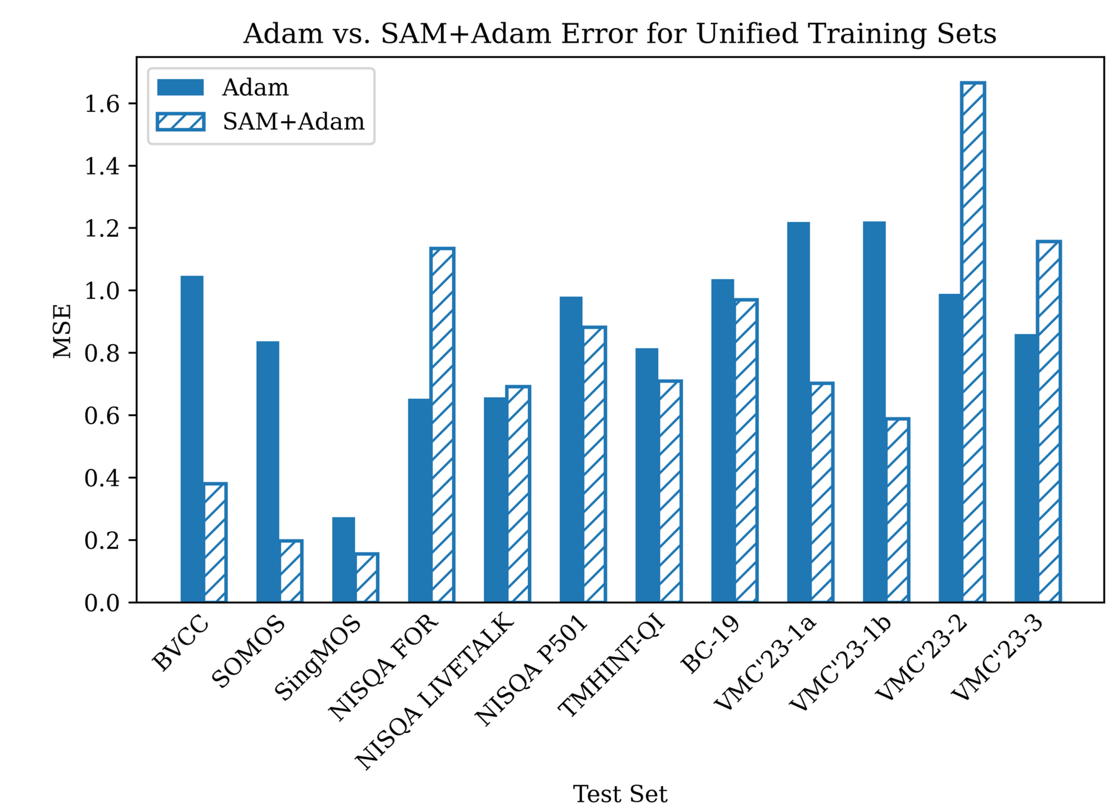


***Fig. 1**: MSE lowers for 8/12 sets with SAM + Adam*

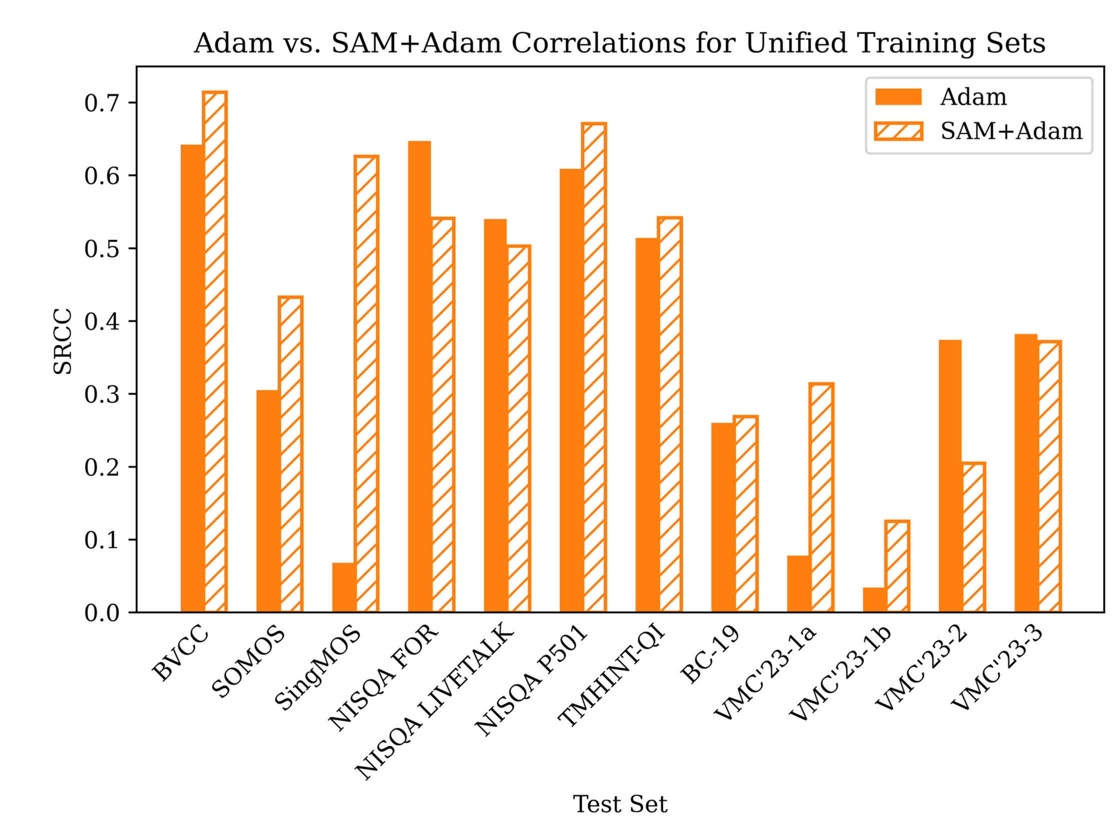


***Fig. 2**: SRCC improves for 8/12 sets with SAM + Adam*

5. KEY TAKEAWAYS

- SAM improves domain generalization.
- Maintains near in-domain performance.
- Provides a simple optimization-based solution with minimal overhead.

Optimization impacts generalization!

SCAN ME
Paper Link

SCAN ME
Portfolio

SCAN ME
Lab Page

THE OHIO STATE UNIVERSITY
COLLEGE OF ENGINEERING